

Combin-AADA

Combin-AADA is a Python project designed to create a HPO (Human Phenotype Ontology) test model using Python, CTWEdge, and CAgen, starting from databases.

Requirements

Before running the scripts, install the following dependencies:

For the Majority of programs operating with csv:

```
pip install pandas
```

For translation:

```
pip install deep_translator
```

For graphics generation:

```
pip install matplotlib
pip install obonet
```

hp.opo is also required.

<https://hpo.jax.org/data/ontology>

Usage

Folder Structure

- The folder contains Python scripts that process and analyze data.
- The input and output CSV files are located in the **output** folder.

Save and Translate

Users who don't need to translate or merge datasets can skip this section.

- **savedatabase.py**: Saves the dataset needed for further operations. Then, it translates a specific dataset (translates 10 rows to English per cycle), and merges the two datasets into a single file.

After running these 3 programs, you will have a single dataset named **merged_dataset**.

Dataset Customization

If you already have the desired dataset, skip this section.

- **dataset_customization.py**: Customizes the dataset and generates two results: one ordered and one unordered.

HPO Hierarchy

- **completehierarchy.py**: Organizes the HPO hierarchy into a CSV file.
- **completehierarchy_from118.py**: Organizes the HPO hierarchy with a maximum parent code of **HP:0000118** into a CSV file.

- **truedepthcalc.py**: Organizes the hierarchy in a table with depth values (depth 0 represents the root, or code **HP:0000001**, while depth 15 is the maximum phenotype specification in HPO).
- **depthfrom118.py**: Organizes the hierarchy into a table with depth values (depth 0 is the new root **HP:0000118**, depth 13 is now the maximum depth).

CTWEdge Input Creation

- **zeroremoval.py**: Removes all zeros from the hierarchy and depth to generate reduced-length CTWEdge files.
- **createctwedgenozero.py**: Creates CTWEdge input files without **HP:+** and all zeros, while preserving hierarchy constraints and handling **NULL** values.

CTWEdge Execution

- Execute CTWEdge on the generated input files and save the result as **code.csv**.

Code Formatting

- **codectwformatting.py**: Formats the codes by converting the abbreviated form **HP:0000001** in the **code.csv** file (input required).

Extraction of Matching Combinations

- **addingparentstodataset.py**: Adds parent codes to their children in the ordered dataset and stores the result in a new column.

Association

- **csvcreator_finale.py**: Matches the codes from the CTWEdge combination with all the corresponding reports in your dataset.
- **find_best_match.py**: Matches the code from the CTWEdge combination with the best corresponding report in your dataset.
- ****from_ctw_to_acts.py****: translate from ctw to acts specifically for ctw hpo

Input & Outputs

Script name	Inputs	Outputs
savedatabase.py		dataset1.csv dataset2.csv
translate10x.py	dataset1.csv	dataset1_translated.csv
mergedataset.py	dataset1_translated.csv dataset2.csv	starting_dataset
dataset_customization.py	starting_dataset.csv	ordered_dataset.csv notordered_dataset.csv
countstartinghpo.py	ordered_dataset.csv	hpo_counts.csv
CountExtension.py	hpo_counts.csv hp.obo	output_hierarchy.csv
completehierarchy.py	hp.obo	completehpohierarchy.csv
completehierarchy_from118.py	hp.obo	completehpohierarchy_from118.csv
truedepthcalc.py	completehpohierarchy.csv	true_depth.csv
depthfrom118.py	completehpohierarchy.csv	true_depth_from118.csv
zeroremoval.py.py	true_depth_from118.csv completehpohierarchy_from118.csv	true_depth_from118_cleaned.csv completehpohierarchy_from118_cleaned.csv

Script name	Inputs	Outputs
createctwedgenozero.py	true_depth_from118_cleaned.csv completehpohierarchy_from118_cleaned.csv	test.ctw
CTWEDGE EXECUTION	test.ctw	codes.csv
codectwformatting.py	code.csv	formatted_code.csv
addingparentstodataset.py	ordered_dataset.csv completehpohierarchy.csv	ordered_dataset_withparents.csv
csvcreator_finale.py	formatted_codes.csv ordered_dataset_withparents.csv	output_data_matches output_data_no_matches output_data_exact_matches
find_best_match.py	formatted_codes.csv ordered_dataset_withparents.csv completehpohierarchy.csv	output_data_matches output_data_no_matches output_data_exact_matches

Contributing

Pull requests are welcome! For major changes, please open an issue first to discuss the proposed changes.

Please ensure to update the tests as appropriate.

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